Spire — SME Validation Checklist

Black-box validation of the Spire framework with an Oracle procurement / HCM subject-matter expert (SME).

Purpose. Validate the framework *as it stands today* — as a black box, without reading code — against what a domain expert knows to be true in Oracle Fusion. If the framework proves correct on the current endpoint set, that result is the basis for widening scope by integrating more endpoints.

Current scope under test. The KG is built over four entities: Supplier, Invoice, PurchaseOrder, and Buyer (cross-pillar, from HCM `publicWorkers`). Supplier is the only valid query target. This checklist tests exactly that surface — not future domains.

How to use. Work top to bottom. For each item, record Pass / Fail / Partial and a note. The goal is to find where the framework is wrong, not to confirm it is right — a failed item found now is worth more than a passed demo.

What Spire is (context for the reviewer)

Spire turns a plain-English procurement question — e.g. "suppliers in the US with open purchase orders and no invoices in six months" — into a correct, structured answer from a live Oracle Fusion pod. It is aimed at consultants and procurement/finance users who shouldn't have to hand-write REST queries across FSCM and HCM.

The design is deliberately bounded: an LLM is used only to interpret the question and phrase the answer; the steps in between — which endpoints to call and which business rules to apply — are deterministic code, so the same question always yields the same result.

The framework's central choice is *who plans the query*. In traditional function-tool calling the LLM is the planner — it picks which tools to call — and this degrades as the tool catalog grows: more tools mean more misrouting and hallucination. Spire moves planning to the KG: the LLM only interprets the question, and tool selection becomes a deterministic KG lookup rather than an LLM judgment. Validating that this KG-driven planning stays correct — and stays correct as scope grows — is what this exercise tests.

What makes this work is the **Knowledge Graph (KG)**: a small authored model of the procurement domain — the entities (Supplier, Invoice, PurchaseOrder, Buyer), how they join, which endpoint serves each, and a **vocabulary** mapping business language to Fusion codes ("vendor" → Supplier, "open" → the PO status code). Validating whether that domain model is *correct* is the heart of this exercise.

Tech stack. LangGraph (pipeline) · KG in Oracle tables · MCP for Fusion calls · provider-swappable LLM (Gemini / OpenAI / Anthropic) · live Oracle Fusion Cloud pod (FSCM + HCM REST).

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1. Answer correctness (the core black-box test)

Give prompts; check outputs against ground truth the SME confirms independently in Fusion (UI or direct REST). Every number should be reconcilable — do not eyeball for plausibility.

#	Item	Method	Result	Note
1.1	Supplier data accuracy — returned suppliers match Fusion	Run a country prompt; SME pulls the same set in Fusion; compare		
1.2	Counts match	Compare record counts		

#	Item	Method	Result	Note
		(returned vs Fusion)		
1.3	Filter correctness — country filter returns the right suppliers	Inspect whether filter matches business meaning, not just a code		
1.4	Join correctness — buyer attached to a PO is the correct buyer	Trace one PO → buyer across FSCM→HCM; SME confirms identity		
1.5	Negation correctness — "no invoices in N months" truly means absence	SME verifies a flagged supplier genuinely has no invoices in range		
1.6	Aggregation correctness — "top N by spend" totals reconcile	Compare rankings + totals against Fusion's own figures		

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2. Domain semantics (where the SME is irreplaceable)

The KG encodes a translation from business language to Fusion codes. Only an SME can confirm that translation is faithful. Validate these from the KG vocabulary/status tables directly — no run needed.

#	Item	Method	Result	Note
2.1	Business-term mapping — "vendor/supplier", "spend authorized", "active" mean what the KG says	Review vocabulary table with SME		
2.2	Status/code fidelity — "open	Confirm the mapping		

#	Item	Method	Result	Note
	PO" maps to the correct StatusCode(s)	is complete (Fusion has more states than OPEN/CLOSED)		
2.3	Country semantics — the country filter uses the business-correct field	Clarify MatchedCountry vs TaxpayerCountryCode: which is "a supplier in X"?		
2.4	Edge-case entities — tax authorities, one-time/ intercompany suppliers	Ask whether these should be counted as "suppliers" in results		

2.3 note: current output shows suppliers with MatchedCountry=IN but TaxpayerCountryCode=US. The SME should rule on whether that is correct for "suppliers in India" — it is exactly the kind of semantic ambiguity only a domain expert settles.

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3. Scope and refusal behavior

#	Item	Method	Result	Note
3.1	Refusal correctness — the right prompts are refused	Feed out-of-scope prompts; SME confirms each is genuinely out of scope		
3.2	No wrongful refusal — in-scope prompts are not rejected	Feed valid but oddly-phrased prompts		
3.3	Silent wrongness —	SME actively hunts for		

#	Item	Method	Result	Note
	confident answers that are subtly wrong	these; the most dangerous failure mode		
3.4	Coverage gaps — common procurement questions the framework can't handle	SME lists real questions a consultant would ask; feeds roadmap		

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4. Trust and honesty

#	Item	Method	Result	Note
4.1	Truncation honesty — partial examination is disclosed	Confirm "N of M examined" appears when source is truncated		
4.2	Hallucination guard — cannot invent a supplier or ID	Try to induce a fabricated entity; confirm it refuses/errors		
4.3	Determinism — same prompt yields the same answer across runs	Run one prompt 2–3 times; compare		

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5. Fitness for real use

#	Item	Method	Result	Note
5.1	Prompt naturalness — works on how consultants	SME writes prompts in their own words,		

#	Item	Method	Result	Note
	really phrase questions	unprompted by examples		
5.2	Answer usefulness — output is in a form a consultant would actually use	SME judges the report/ answer format		
5.3	Explanation legibility — a consultant can tell what the system did	Review the intent/plan/ answer trace for clarity		

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6. Forward-looking: readiness to widen scope

Only relevant *if* sections A–E pass. This is the "what next" the SME helps shape — not part of the current framework, but the plan it unlocks.

#	Item	Method	Result	Note
6.1	Endpoint priority — which new Fusion endpoints add the most value next	SME ranks candidate domains by real consultant demand		
6.2	Endpoint truth — for each new endpoint, the real join keys and filterable fields	SME confirms the field/key facts before KG authoring		
6.3	Access/security model — what a new endpoint requires (scope grants, self-service vs admin)	SME describes how that data is governed in practice		
6.4	Semantic pitfalls — known	SME flags the		

#	Item	Method	Result	Note
	asymmetries or quirks in candidate endpoints	"SupplierId vs SupplierNumber"-style traps in advance		

Materials to give the SME

1. **A prompt-to-output test set** — the prompts you will run, so the SME can check answers against Fusion ground truth.
2. **The KG vocabulary + status mappings as a plain table** — so the SME
   can validate semantics (section B) directly, without running anything.
3. **The architecture explainer** — one read so they understand what the black box claims to do before probing it.

The one rule for the "report" validation

A report that *looks* right can still be built on wrong data. Every figure in the report must be independently reconcilable against Fusion — the SME pulls the same query in Fusion and compares. Plausibility is not validation. (This is the Phase-A lesson: a misread was only caught by a direct call against the API, not by inspection.)

Appendix — Validation prompt set

A representative set to run during validation, grouped by expected outcome. The right-hand column links each prompt to the check item(s) it

exercises. A prompt that behaves outside its group (e.g. a refusal that returns data) is an immediate fail.

Supported — must return data

Prompt	Exercises
List all suppliers in India	1.1, 1.2, 2.3
Show me suppliers in the United Kingdom	1.1, 1.3
Find suppliers in US	1.3, 2.3
List active suppliers in the US	1.3, 2.1
US suppliers with no invoices in the last 6 months	1.5
Suppliers in the US with open purchase orders and no invoices in 6 months, include the buyer	1.4, 1.5, 2.2
Which US vendors have open POs? Show the purchasing agent.	1.4, 2.1, 2.2

Aggregation — must return rankings

Prompt	Exercises
Top 10 suppliers by spend in 2022	1.6
Which suppliers did we spend the most with in 2022	1.6
Show me the top 5 suppliers by total invoice amount in 2022	1.6

Refusals — must be declined (out of scope)

Prompt	Exercises
Show me invoices for supplier 1252	3.1

Prompt	Exercises
Show me contract renewals due next quarter	3.1
Top 10 suppliers by total spend (no date range)	3.1
Who is our best supplier?	3.1

Suggested SME-authored prompts

Ask the SME to add 3–5 prompts in their own words, unprompted by the examples above — this tests prompt naturalness (5.1) and surfaces coverage gaps (3.4). Record each with its outcome and whether it was correct.

SME prompt	Expected	Outcome	Correct?